



FEET AND CALVES

The skeletal structure of the lower leg and foot is made up of the long tibia and fibula bones found in the lower leg and the small foot bones called tarsals, metatarsals, and phalanges. These bones form numerous joints. The most important is the ankle joint, located between the tibia bone of the leg and the talus of the foot. This is a hinge joint, and it is involved with the major joint movements of plantar flexion (toes point down) and dorsiflexion (toes point up).

The other major joints found between each of the tarsal and metatarsal bones are gliding joints. They allow more limited movements of the foot. When several of these gliding joints are working together in the foot, a much broader range of movement is achieved compared with the movement produced by a single gliding joint working alone. Thus, multiple-joint movements allow for foot eversion (sole turned out) and inversion (sole turned in).

The joints that allow the most freedom of movement of the foot are the condyloid joints, located between the metatarsal bones and the phalanges. Condyloid joints allow the movements of flexion, extension, abduction, adduction, and circumduction of the toes. Finally, the joints that allow for flexion and extension of the toes are the hinge joints between the phalanges.

Without the ligaments and connective tissues found in the lower leg and foot, joint movement and muscle function would be greatly compromised. The joints in the foot are connected to each other by many ligaments. The largest ligament in this area is the deltoid ligament, or ankle medial collateral ligament. It is composed of four segments that connect the tibia to the talus, calcaneus, and navicular bones. Opposite the deltoid ligament is the ankle lateral collateral ligament, which is composed of three segments that connect the fibula to the talus and calcaneus bones. Because the deltoid ligament is much stronger than the ankle lateral collateral ligament, and the tibia is longer than the fibula, the ankle is predisposed toward inversion (turning in).

Retinacula are another type of connective tissue located in the lower leg that secure many of the muscle–tendon units. This support allows these muscles to work harder, stronger, and more efficiently. The superior and inferior retinacula in the dorsal (top) area of the foot hold down all the tendons of the extensor muscles. On the lower lateral side of the foot, the peroneal retinaculum holds down the tendons of the peroneus longus and peroneus brevis

muscles. The flexor retinaculum on the medial side of the ankle holds down the tendons of the flexor digitorum longus, flexor hallucis longus, and tibialis posterior muscles. These muscles are illustrated on page 16.

The final noteworthy connective tissue is the plantar fascia. The plantar fascia is a broad, thick connective tissue that supports the arch on the bottom of the foot. It spans the area between the tuberosity of the calcaneus and the heads of the metatarsal bones.

The muscles that move the ankle and toes are located primarily in the lower leg (figure 2.1); these muscles have tendons that are as long as or longer than the muscles. The dominant tendon is the Achilles tendon, which is shared by the gastrocnemius, plantaris, and soleus. The gastrocnemius and soleus muscles are the prime plantar flexors and are assisted by the plantaris and tibialis posterior as well as two toe flexor muscles, flexor digitorum longus and flexor hallucis longus. Located on the outer (lateral) side of the calf is another group of three muscles—peroneus longus, peroneus brevis, and peroneus tertius—which are used in evverting the foot. Additionally, the peroneus longus and peroneus brevis plantar flex the ankle.

Three anterior calf muscles (tibialis anterior, extensor hallucis longus, and extensor digitorum longus) dorsiflex the ankle as well as move the foot and toes. The extensor digitorum brevis, dorsal interosseous, and extensor hallucis brevis muscles are located on the dorsal (top) side of the foot and extend the toes.

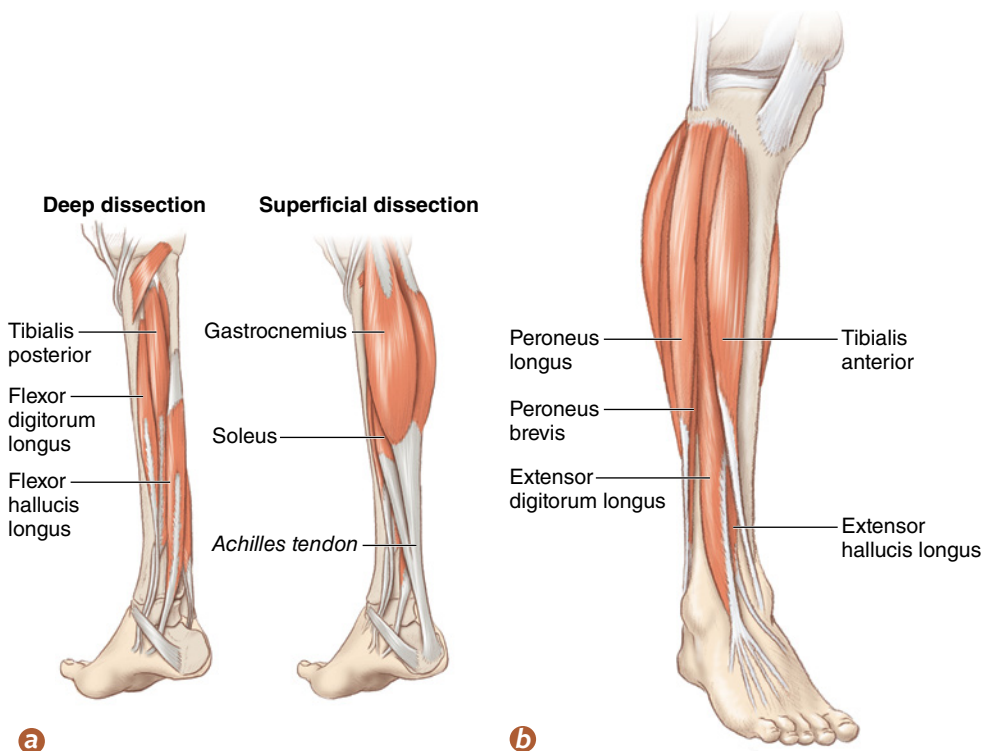


FIGURE 2.1 Calf and foot muscles: (a) posterior; (b) anterior.

The muscles on the plantar (sole) side of the foot, the flexor digitorum brevis, quadratus plantae, flexor hallucis brevis, flexor digiti minimi brevis, abductor hallucis, abductor digiti minimi, plantar interosseous, and lumbricals, are used to flex and spread the toes.

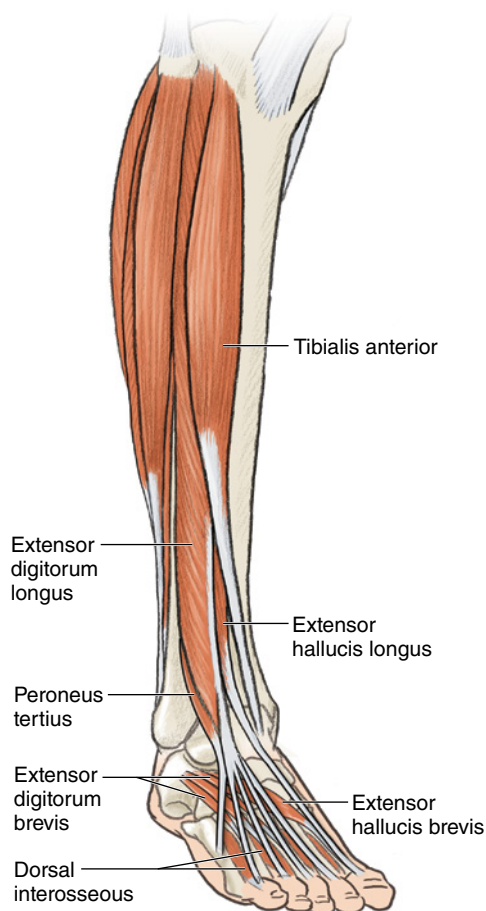
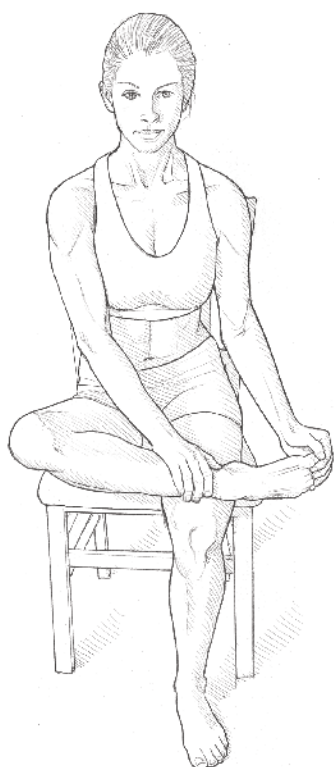
The movement range for the ankle and toes is limited by the strength of the agonist muscles, flexibility of the antagonist muscles, tightness of the ligaments, and bone contacts or impingements. One of the most notable limiters is the plantar fascia. A tight plantar fascia limits toe extension, and in cases where the fascia is inflamed, it will also limit plantar flexion. The range of motion for both plantar flexion and dorsiflexion can also be limited by the formation of bone spurs. Excessive stress can activate bone cells to form bone spurs on the anterior and posterior lips of the talus and the superior tibial dorsal neck. These bony outcroppings cause more rapid bone contacts, thus ending the movement. Interestingly, most of the range-of-motion limiters, except bone impingements, can be changed by doing stretching exercises.

People use the muscles of their feet and lower legs extensively during normal daily activities such as standing in line and walking from place to place. Although the musculature of the lower leg is substantially smaller than that of the upper leg, it supports the entire body and receives the heaviest load during these activities. Because the feet are also constantly exerting force against whatever surface they are in contact with, it is no wonder that many people end up with minor aches, cramps, and weakness in the muscles of the lower legs and feet toward the end of the day. Stretching and strengthening these smaller muscle groups can alleviate much of the fatigue and pain caused by daily activities. In addition to helping reduce pain, stretching the muscles of the lower leg and foot can also improve overall flexibility, strength, strength endurance, balance, and stamina. Improving strength and flexibility in these muscle groups generally will enable people to be more productive by increasing their ability to work longer and harder at their jobs or during recreation activities.

Pain, cramping, restlessness, and weakness in the arch of the foot and calf muscles are common complaints. Problems such as these often result from the continual heavy loads put on the muscles. Chronic use of these muscles can also increase muscle tightness. Tightness may lead to conditions such as tendinitis and shin splints. Achilles tendinitis, associated with overuse and tightness of the gastrocnemius and soleus muscles, is quite common, in fact. Shin splints result from inflammation of the frontal compartment of the lower-leg muscles, the tibialis anterior and, in some cases, the soleus and flexor digitorum longus. Either of these conditions can become excruciating if not treated in the early stages. A variety of stretching and strengthening exercises for those muscle groups will, in most cases, improve these conditions and help prevent future episodes.

Another common condition is delayed-onset muscle soreness, or DOMS. This problem typically occurs after people participate in unusual or unfamiliar activities. The calf muscles tend to be affected by DOMS more often than any other muscle group in the body. Light stretching exercises are recommended to help improve this condition and relieve some of the pain associated with it.

BEGINNER SEATED TOE EXTENSOR STRETCH



Execution

1. While sitting on a chair with the left foot on the floor, raise the right ankle and place it on top of the left knee.
2. While bracing the right ankle with the right hand, place the fingers of the left hand on the tops of the right toes.
3. Pull the tips of the toes toward the sole of the foot, away from the tibia bone.
4. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right extensor digitorum longus, right extensor digitorum brevis, right extensor hallucis longus, right extensor hallucis brevis, right tibialis anterior, right peroneus tertius

Less-stretched muscle: Right dorsal interosseous

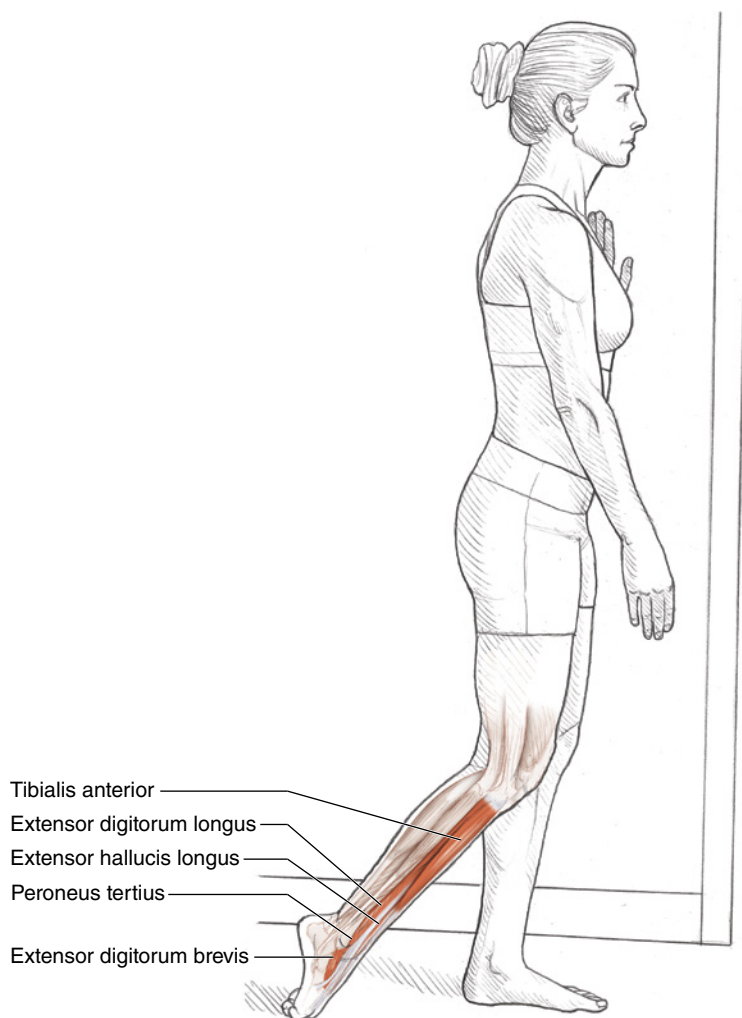
Stretch Notes

This stretch alleviates minor aches and tightness in the toe extensor muscles located on the top of the foot. Generally speaking, these muscles are not as strong as the toe flexor muscles located on the bottom of the foot because they are not exerting force against the ground in daily activities such as running and walking. Rather, they are constantly used as antagonist muscles in clearing the ground (toe extension and dorsiflexion) while walking or running. Consequently, they tend to become less sore or stiff than toe flexor muscles.

This stretch is one of the easiest to execute. It can be done while you sit watching TV or during a similar seated activity. When you are relaxing at the end of the day, regular stretching of these muscles will do wonders. A morning stretching routine is also a beneficial way to start the day. The series of stretching exercises can be done any time during the day.

Hold the ankle firmly in order to keep it and the foot stable. You will feel the stretch on the top of the foot (dorsal side). If grasping and pulling on the tips of the toes causes too much pain, apply the pressure at the ball of the foot.

ADVANCED STANDING TOE EXTENSOR STRETCH



Execution

1. Stand upright and brace against a wall or an object for balance.
2. Point the right foot backward away from the body, with the dorsal (top) side of the toes down against the floor. Placing the top side of the foot on a pillow or towel makes this stretch more comfortable.
3. While keeping the dorsal side of the toes pressed against the floor, lean your weight onto the right leg, and press the bottom of the heel down toward the floor.
4. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right extensor digitorum longus, right extensor digitorum brevis, right extensor hallucis longus, right extensor hallucis brevis, right tibialis anterior, right peroneus tertius

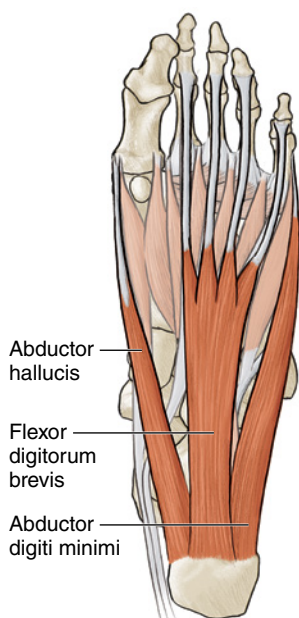
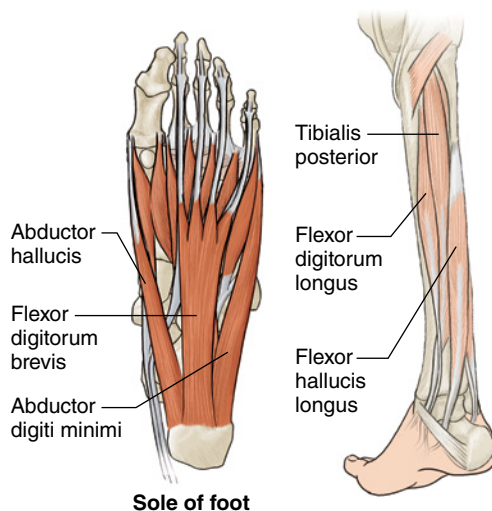
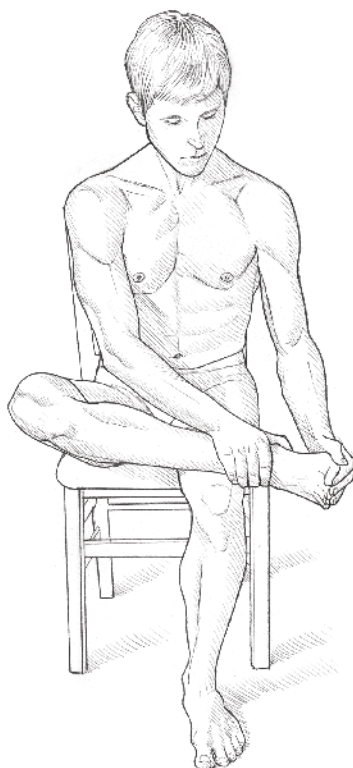
Less-stretched muscle: Right dorsal interosseous

Stretch Notes

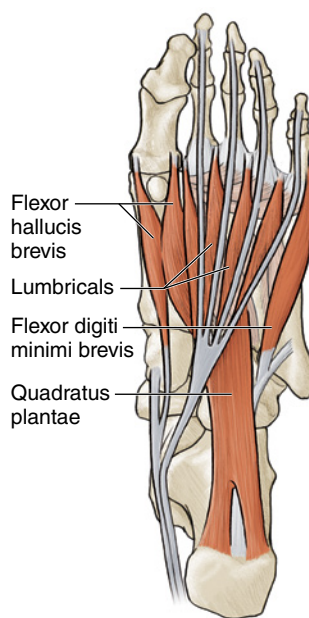
Many fitness exercisers experience shin splints in front of the tibia bone. This condition is very painful during exercise. This condition is associated with inflammation of the tibialis anterior muscle and the connective tissue around the anterior compartment of the tibia. It is often caused by overuse or tightness of the tibialis anterior muscle. It can also be associated with the type of shoes you wear and the surfaces on which you exercise. People with shin splint problems definitely will benefit from this stretch. Shoes and running and walking surfaces also need to be evaluated.

It is more comfortable to perform this stretch on a carpet or other soft surface, or you can put a pillow or towel between the top of the foot and the floor. Do not drag the foot that is pressed to the floor. Moving the heel medially or laterally will place greater stretch on either the dorsal medial (inner) or dorsal lateral (outer) parts of the foot. It is also recommended that you explore each of these stretches from different angles of pull. This way you are able to locate the sore spots or tightness in these muscles. This stretch is more effective than the previous one. In this stretch, the weight of your whole body puts more stress on these muscles as you stretch.

BEGINNER SEATED TOE FLEXOR STRETCH



Superficial dissection



Intermediate dissection

Execution

1. While sitting on a chair with the left foot on the floor, raise the right ankle and place it on top of the left knee.
2. Brace the right ankle with the right hand, and place the fingers of the left hand along the bottoms of the toes of the right foot, with the fingers pointing in the same direction as the toes.
3. Use the fingers of the left hand to push the toes of the right foot toward the right knee.
4. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right flexor digitorum brevis, right quadratus plantae, right flexor digiti minimi brevis, right flexor hallucis brevis, right lumbricals, right plantar interosseous, right abductor hallucis, right abductor digiti minimi

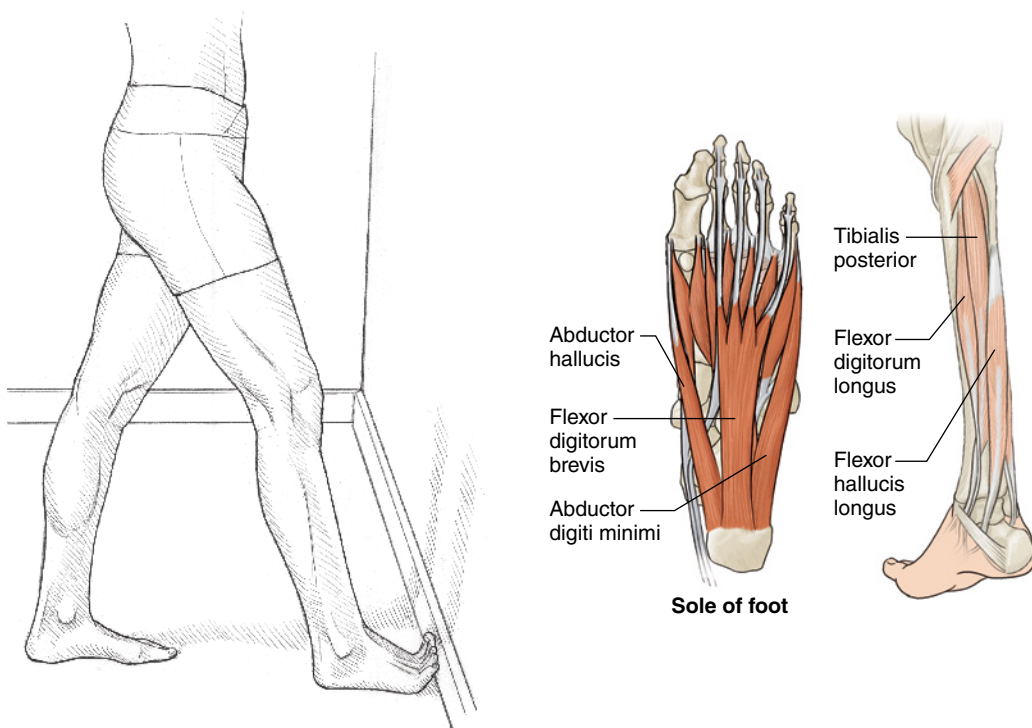
Less-stretched muscles: Right flexor digitorum longus, right flexor hallucis longus, right tibialis posterior, right peroneus longus, right peroneus brevis, right plantaris, right soleus, right gastrocnemius

Stretch Notes

The muscles located in the arch of the foot receive constant stress during daily activities. The stress is caused by supporting body weight during activities such as standing, walking, jumping, and running. Toe muscles also apply force against the ground whenever you are moving. Thus they are in constant use for most of the day, especially if you are an active person. After walking and standing long hours, foot muscles are often more tired, sore, and tight than any other muscle group in the body. You might even experience cramping of these muscles after a long day's work. Stretching these toe flexor muscles will help reduce the pain and soreness after a hard day's work and make you feel better. The muscles of the bottom of the foot are quite sensitive and respond to stretching exercises very well. Light massage along with light stretching exercises make you feel pleasantly relaxed after you have been on your feet most of the day.

Stabilize the foot and ankle with a firm hold. Pushing hard on the very ends of the toes with the left palm will provide a much greater stretch. You will feel the stretch on the sole (plantar side) of the foot.

ADVANCED STANDING TOE FLEXOR STRETCH



Execution

1. Stand upright while facing a wall, 1 to 2 feet (30 to 60 cm) away.
2. Keeping the heel of the right foot on the floor, press the bottoms of the toes of the right foot against the wall. The ball of the foot should be more than half an inch (more than 2 cm) above the floor.
3. Lean forward and slide the ball of the right foot slowly down, keeping the toes pressed against the wall.
4. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right flexor digitorum brevis, right quadratus plantae, right flexor digiti minimi brevis, right flexor hallucis brevis, right lumbricals, right plantar interosseous, right abductor hallucis, right abductor digiti minimi

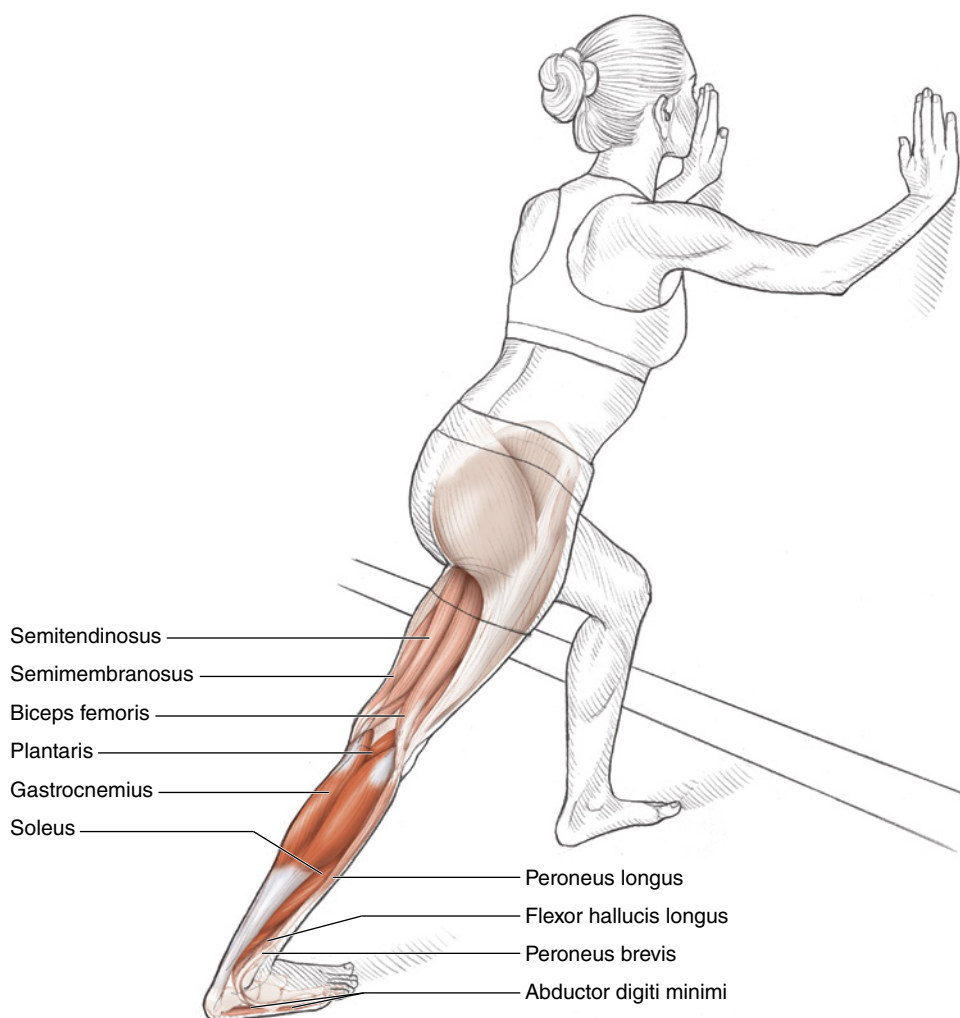
Less-stretched muscles: Right flexor digitorum longus, right flexor hallucis longus, right tibialis posterior

Stretch Notes

Have you driven a car for many hours without stopping? Have you ever felt that your foot is getting tired of moving the gas pedal up and down or being held in the same place for a long time? It happens to most of us. The muscles of the foot are not used to this. They simply get tired. This stretch or any of the previous stretches would be beneficial during a long drive.

Make sure the ball of the foot is parallel to the floor. This ensures that all the toes are stretched equally. Also, slide the ball of the foot down slowly to prevent overstretching. Bending the right knee slightly and moving the knee forward toward the wall will incorporate the calf muscles into the stretch.

BEGINNER PLANTAR FLEXOR STRETCH



Execution

1. Stand facing a wall and brace your hands against it.
2. Place the left foot 1 to 2 feet (30 to 60 cm) away from the wall and the right foot 2 to 4 feet (60 to 120 cm) from the wall.
3. Keeping the right heel on the floor, lean your chest toward the wall. You can bend the left knee slightly to facilitate moving the chest against the wall.
4. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right gastrocnemius, right soleus, right plantaris, right popliteus, right flexor digitorum longus, right flexor hallucis longus, right tibialis posterior

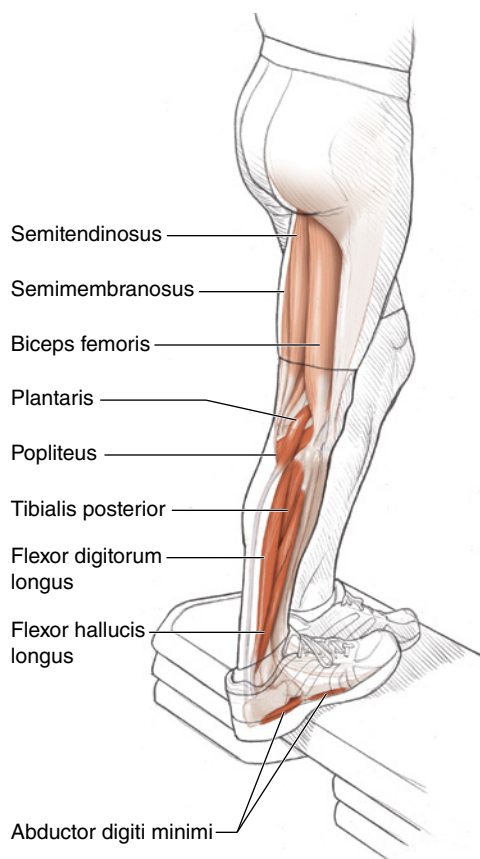
Less-stretched muscles: Right peroneus longus, right peroneus brevis, right flexor digitorum brevis, right quadratus plantae, right flexor digiti minimi brevis, right flexor hallucis brevis, right abductor digiti minimi, right abductor hallucis, right hamstring (semitendinosus, semimembranosus, biceps femoris)

Stretch Notes

Any time you start a new exercise program or participate in unusual or unfamiliar activities, you might experience muscular soreness during the days that follow. This is commonly known as delayed-onset muscle soreness, or DOMS. This painful feeling is felt most often 24 to 72 hours after the exercise. Walking or running uphill or downhill typically produces this painful effect. Calf muscles are usually affected more than any other muscle group in the body. Repeated stretching of these muscles for several days will help relieve the pain of DOMS.

As the chest gets closer to the wall, bending the knee slightly will realign the tibia and increase the distance between the muscle attachment points. This will increase the stretch on the tibialis posterior, flexor hallucis longus, and flexor digitorum longus muscles while at the same time reducing the stretch on the hamstring muscles.

ADVANCED PLANTAR FLEXOR STRETCH



Execution

1. Stand upright on the edge of a step or beam, with the midsection of the right foot on the edge. Hold on to a support with at least one hand.
2. Keep the right knee straight and the left knee slightly bent.
3. Lower the right heel as far as possible.
4. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right gastrocnemius, right soleus, right plantaris, right popliteus, right flexor digitorum longus, right flexor digitorum brevis, right flexor hallucis longus, right flexor hallucis brevis, right tibialis posterior, right quadratus plantae, right flexor digiti minimi brevis, right abductor digiti minimi, right abductor hallucis

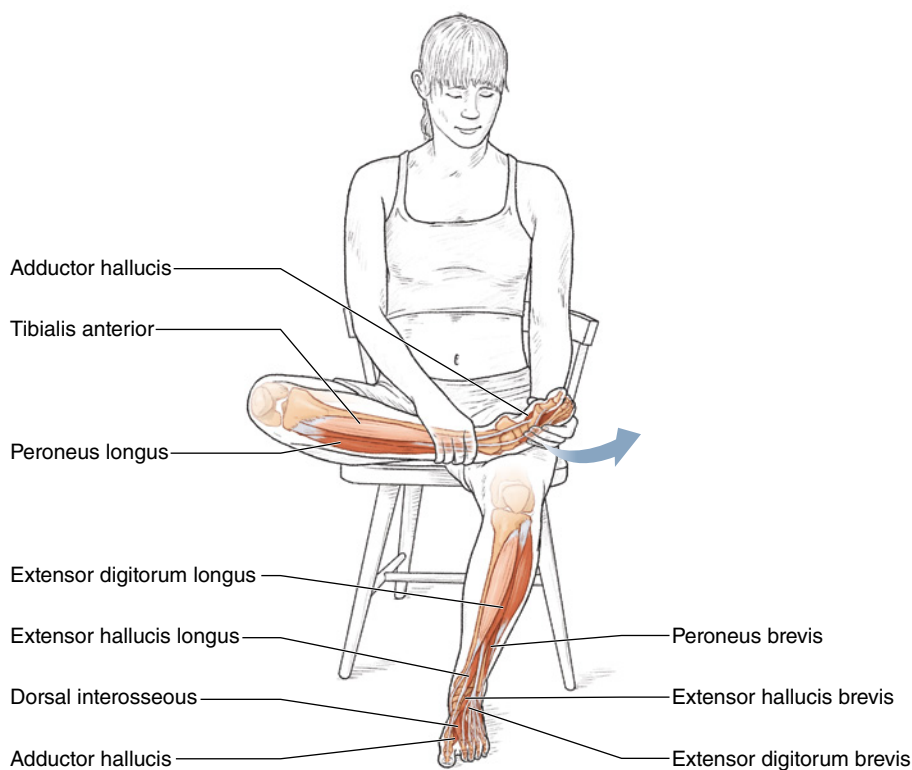
Less-stretched muscles: Right hamstring (semitendinosus, semimembranosus, biceps femoris)

Stretch Notes

Many recreational and even competitive runners experience a condition called tendinitis, the chronic inflammation of a tendon. Tendinitis is caused by chronic overuse and tightness of the muscles associated with the affected tendons. The most vulnerable place for this condition in the lower leg is the Achilles tendon. The gastrocnemius and soleus muscles attach to this tendon. If not treated, tendinitis of the Achilles will become excruciatingly painful and will limit your participation in almost any sport activity. Research shows that most people simply don't spend enough time and effort stretching these muscles. Often it takes a long time (perhaps months) to get rid of this tendinitis. A good stretching program for these muscles should be part of your overall training program.

This stretch is the best for your calf muscles in general. It is more comfortable to do this stretch while wearing shoes. Always support the body. Not supporting the body could cause the muscles to contract and not stretch. After the heels reach their lowest point, apply more stretch by bending the knees slightly. This will stretch the tibialis posterior, flexor hallucis longus, and flexor digitorum longus muscles while reducing the stretch on the hamstring muscles. Placing the ball of the foot on the edge of the stairs or beam will increase the stretch on the origin (top part) of these muscle groups. Placing the midsection of the foot on the edge of the stairs or beam increases the stretch on the lower portion of these muscles. The sharper the edge of the stair, the better grip you are able to produce between the stair and the foot, and the more stretch you are able to produce on these muscles.

BEGINNER SEATED TOE EXTENSOR AND FOOT EVERTOR STRETCH



Execution

1. While sitting on a chair with the left foot on the floor, raise the right ankle and place it on top of the left knee.
2. While bracing the right ankle with the right hand, place the thumb of the left hand along the ball of the right big toe and place the fingers of the left hand across the top of the right foot.
3. Pull the tips of the toes toward the sole of the foot, away from the tibia bone.
4. Simultaneously, while keeping the ankle braced, pull and slightly twist and rotate the foot steadily upward.
5. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right abductor digiti minimi, right peroneus longus, right peroneus brevis, right peroneus tertius, right extensor digitorum longus, right extensor digitorum brevis, right extensor hallucis longus, right extensor hallucis brevis, right adductor hallucis.

Less-stretched muscle: Right dorsal interosseous, right tibialis anterior

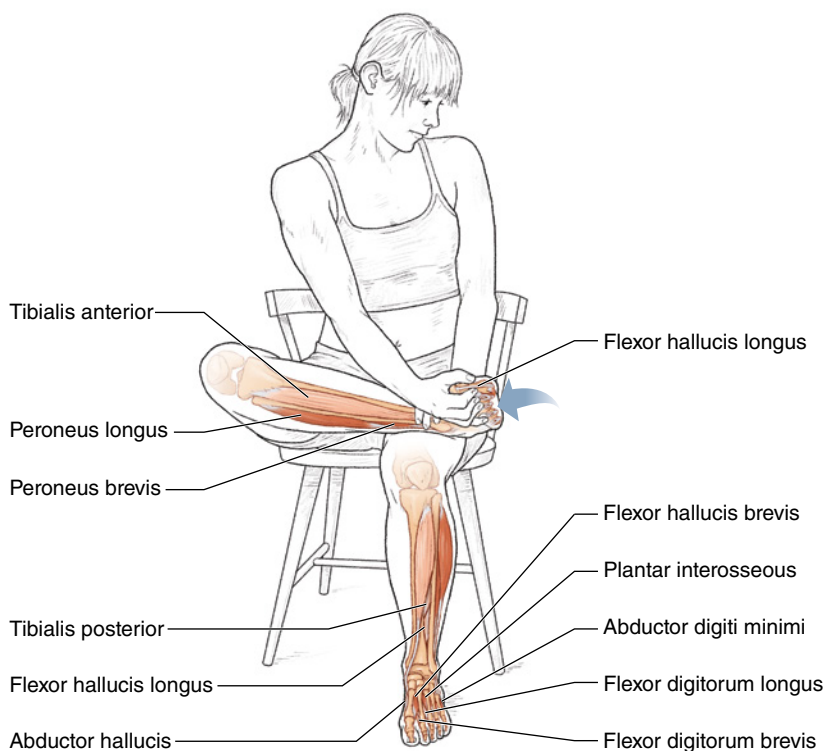
Stretch Notes

This is a good beginner's stretch to alleviate minor aches and tightness in the toe extensor muscles located on the top of the foot as well as the foot evor muscles located on the lateral side of the foot and the calf. In this stretch, you should feel an equal amount of stretch of the toe extensor muscles and the foot evor muscles. The abductor digiti minimi, the three peroneus muscles, and the extensor hallucis brevis will experience extensive stretch during this exercise. In addition, this will stretch important ligaments located on the dorsal and lateral sides of the foot.

This stretch and other foot and ankle stretches are easy to execute at a variety of times and places throughout the day. You can perform this stretch while you are working in the office, taking a break at work, relaxing at home, and just about anywhere you are sitting.

When executing this stretch, stabilize the foot and ankle with a firm hold during the entire stretch. Grasping the ends of the toes, pulling, and slightly rotating them upward (while keeping the toes in the extended position) can produce a more effective stretch.

BEGINNER SEATED TOE FLEXOR AND FOOT EVERTOR STRETCH



Execution

1. While sitting on a chair with the left foot on the floor, raise the right ankle and place it on top of the left knee.
2. While bracing the right ankle with the right hand, place the thumb of the left hand on the top of the right big toe and the fingers of the left hand across the top of the toes on the right foot.
3. Pull the tips of the toes away from the sole of the foot toward the tibia bone.
4. Simultaneously, while keeping the ankle braced, pull and slightly twist and rotate the foot steadily upward.
5. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right peroneus longus, right peroneus brevis, right abductor digiti minimi, right flexor digitorum brevis, right quadratus plantae, right flexor digiti minimi brevis, right flexor hallucis brevis, right lumbricals, right plantar interosseous

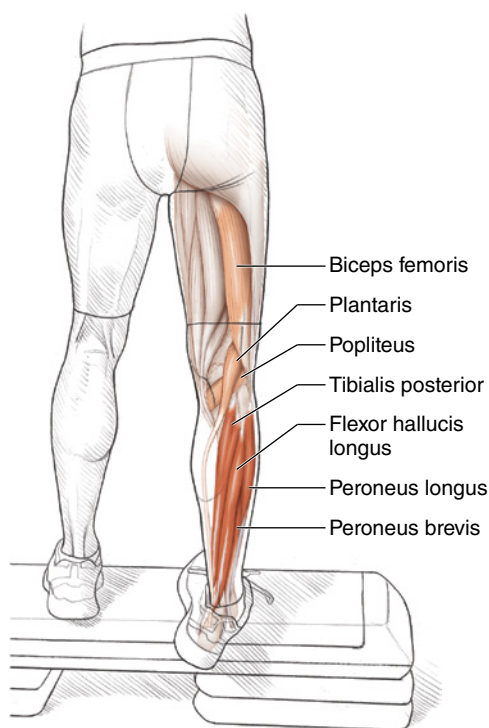
Less-stretched muscles: Right flexor digitorum longus, right flexor hallucis longus, right tibialis posterior, right plantaris, right soleus, right gastrocnemius

Stretch Notes

The muscles of the foot, ankle, and calf are in constant use throughout the day. Muscles associated with these parts of the body need extra care and attention. Stretching and light massage of these muscles bring great relief and even pleasure if the muscles are taken care of daily. Stretching these toe flexor and foot evtor muscles will help reduce the pain and soreness after a hard day's work and make you feel better. The muscles of the bottom of the foot are sensitive and respond well to stretching exercises. Light massage along with light stretching exercises make you feel pleasantly relaxed after you have been on your feet most of the day.

Stabilize the foot and ankle with a firm hold. Pulling and twisting the ends of the toes with the left fingers and thumb provide a greater stretch. You will feel the muscle pull on the sole of the foot and on the lateral side of the foot, ankle, and calf. In addition, this exercise will stretch important ligaments located on the plantar (sole) side and the lateral side of the foot.

ADVANCED PLANTAR FLEXOR AND FOOT EVERTOR STRETCH



Execution

1. Stand upright on the edge of a step or beam, with the midsection of the right foot on the edge.
2. Place the foot in an inverted position by standing on the lateral (outer) side of the foot.
3. Keep the right knee straight and the left knee slightly bent.
4. Hold on to a support with at least one hand.
5. Keeping the foot inverted, lower the right heel as far as possible.
6. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right peroneus longus, right peroneus brevis, right peroneus tertius, right abductor digiti minimi, lateral side of right soleus, lateral side of right gastrocnemius, right flexor hallucis longus, right tibialis posterior

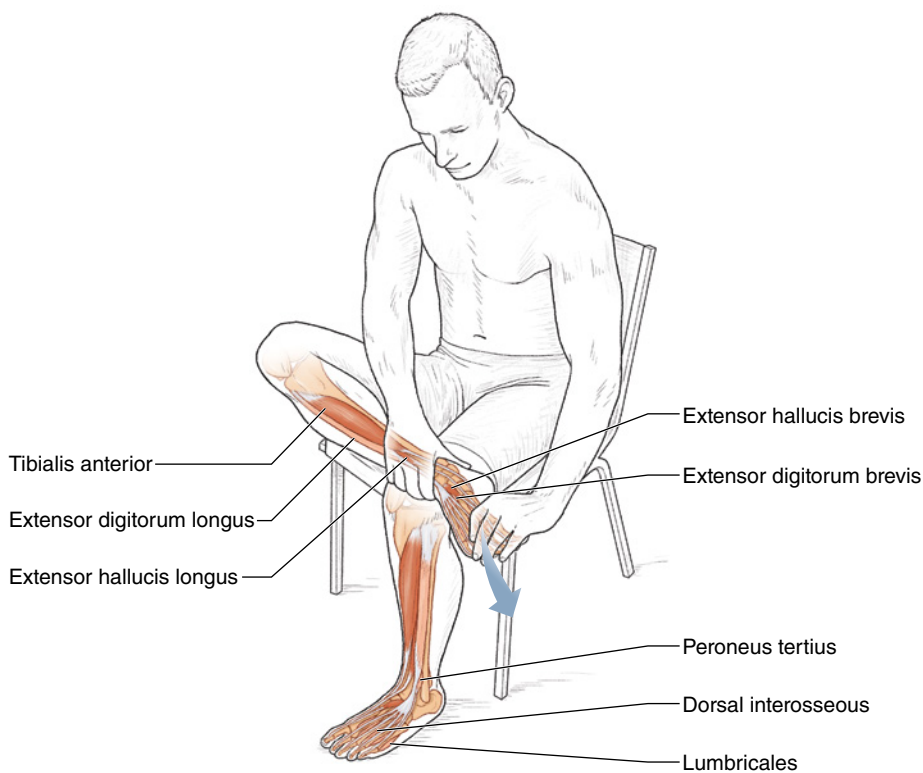
Less-stretched muscles: Right popliteus, right plantaris, medial head of right gastrocnemius, right biceps femoris, right flexor digitorum brevis, right quadratus plantae, right flexor digiti minimi brevis, right flexor hallucis brevis

Stretch Notes

Once in a while many of us experience soreness and tightness on the lateral side (outer side) of the calf muscles. This could happen any time we walk or run on an uneven or unstable surface, such as grass or beach sand, or walk or run downhill. Often this soreness is felt on the days after the activity. This condition is called delayed-onset muscle soreness, or DOMS. When you encounter this problem, it is highly recommended to start a stretching program, especially of the muscles where this pain is felt. This particular stretch is helpful for the lateral (outer) side of the lower leg.

It is more comfortable to do this stretch while wearing shoes. This is an excellent stretch for the peroneus longus and peroneus brevis and the abductor digiti minimi muscles, which are located on the lateral (outer) side of the lower leg and the foot. Be extra careful when placing the foot in an inverted position, and be sure to progress slowly through this stretching exercise. After the right heel reaches the floor or the lowest point, increase the stretch by bending the right knee slightly. This removes any stretch on the hamstring muscles, but it further stretches the calf muscles.

BEGINNER SEATED TOE EXTENSOR AND FOOT INVERTOR STRETCH



Execution

1. While sitting on a chair with the left foot on the floor, raise the right ankle and place it on top of the left knee.
2. While bracing the right ankle with the right hand, place the thumb of the left hand along the ball of the right big toe and the fingers of the left hand across the top of the right foot.
3. Pull the tips of the toes toward the sole of the foot and away from the tibia bone.
4. Simultaneously, while keeping the ankle braced, push and slightly twist or rotate the foot in a steady downward direction.
5. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right abductor hallucis, right tibialis anterior, right extensor digitorum longus, right extensor digitorum brevis, right extensor hallucis longus, right extensor hallucis brevis.

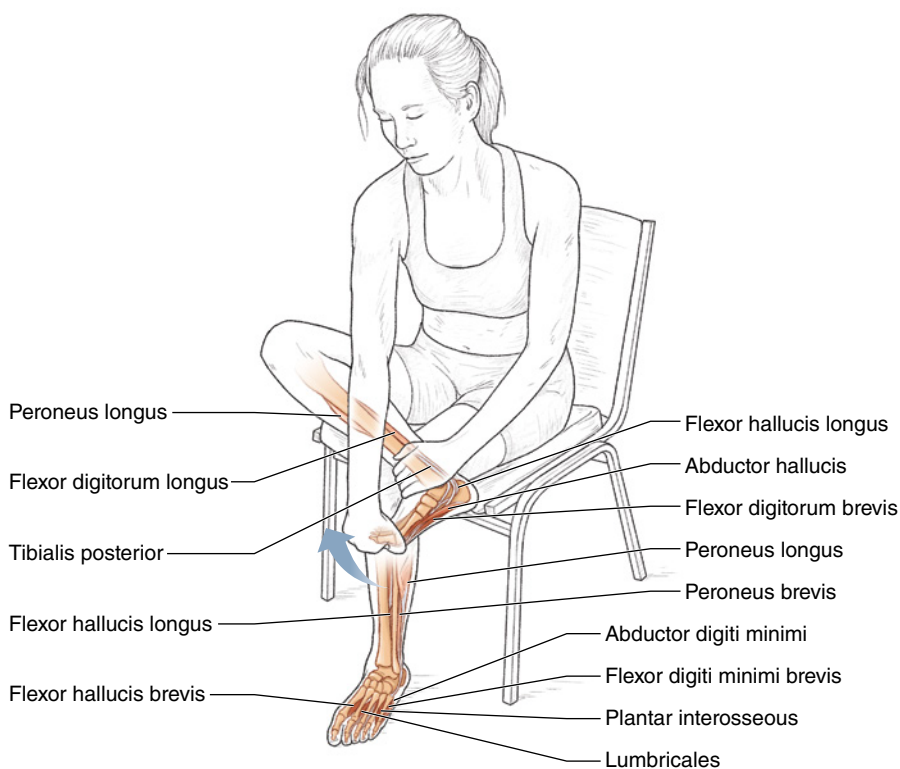
Less-stretched muscle: Right dorsal interosseous, right tibialis posterior, right flexor hallucis longus, right flexor digitorum longus

Stretch Notes

This stretch is good for beginners. It will help alleviate minor aches and tightness in the toe extensor muscles located on the top of the foot and also foot invertor muscles located on the medial side of the foot and the calf. In this stretch, you will feel more muscle stretch on the dorsal side of the foot and a lesser degree of pull on the medial side of the foot (big toe side), ankle, and calf areas. The abductor hallucis, extensor digitorum brevis, and extensor hallucis brevis will experience extensive stretch during this routine. In addition, this routine will stretch important ligaments located on the dorsal and medial sides of the foot.

While executing this stretch, stabilize the foot and ankle with a firm hold during the entire stretch. Grasping the ends of the toes and pushing and slightly rotating them downward (while keeping the toes in the extended position) can produce the desired results from this stretch.

BEGINNER SEATED TOE FLEXOR AND FOOT INVERTOR STRETCH



Execution

1. While sitting on a chair with the left foot on the floor, raise the right ankle and place it on top of the left knee.
2. While bracing the right ankle with the left hand, place the thumb of the right hand on the top of the right big toe and the fingers of the right hand across the bottoms of the toes of the right foot.
3. Pull the tips of the toes toward the tibia bone.
4. Simultaneously, while keeping the ankle braced, push and slightly twist and rotate the foot steadily downward.
5. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right flexor digitorum brevis, right quadratus plantae, right flexor digiti minimi brevis, right flexor hallucis brevis, right lumbricals, right plantar interosseous, right abductor hallucis, right abductor digiti minimi

Less-stretched muscles: Right flexor digitorum longus, right flexor hallucis longus, right tibialis posterior, right peroneus longus, right peroneus brevis, right plantaris, right soleus, right gastrocnemius

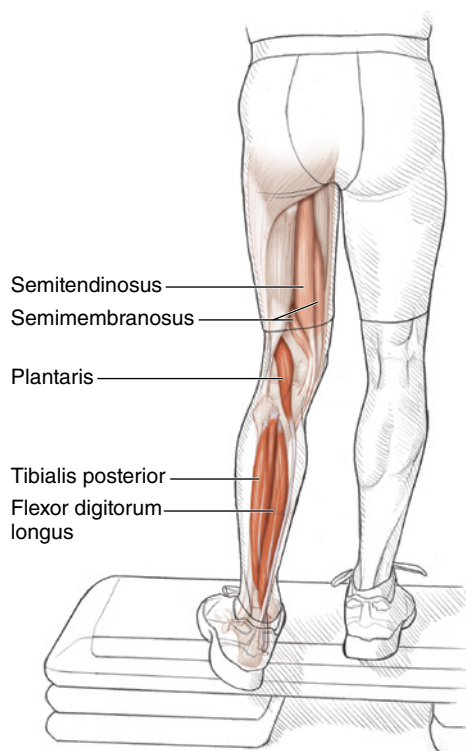
Stretch Notes

Constant stress on foot, ankle, and calf muscles often causes these muscles to become tight and stiff. This in turn, can lead to chronic muscle soreness or injury because of the tendency to favor one side of the body over the other while walking or running. One of the best ways to prevent this is to start a regular stretching program.

The muscles of the foot, ankle, and lower leg are sensitive and respond well to stretching exercises. All foot and lower-leg stretches are easy to execute during the day while in a seated position. These stretches done in 5- to 10-minute segments can be accomplished repeatedly throughout the day. Light massage along with light stretching exercises make you feel pleasantly relaxed after you have been on your feet most of the day.

In this stretch, stabilize the foot and ankle with a firm hold. Pushing on and rotating the ends of the toes with the left fingers and thumb will provide a greater stretch. You will feel the stretch on the sole of the foot and on the medial side of the foot, ankle, and calf. In addition, this will stretch important ligaments located on the plantar (sole) and medial side of the foot.

ADVANCED PLANTAR FLEXOR AND FOOT INVERTOR STRETCH



Execution

1. Stand upright on the edge of a step or beam, with the midsection of the left foot on the edge.
2. Place the foot in an everted position by standing on the medial (inner) side of the foot.
3. Bend the left knee slightly toward the midsection of the body (inside direction), with the right knee slightly bent.
4. Hold on to a support with at least one hand.
5. While keeping the foot everted, lower the left heel as far as possible.
6. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Left flexor digitorum longus, left abductor hallucis, medial side of left soleus, left tibialis posterior, left plantaris

Less-stretched muscles: Left flexor digitorum brevis, left quadratus plantae, left flexor hallucis brevis, left flexor digiti minimi brevis, left medial gastrocnemius, left semitendinosus, left semimembranosus

Stretch Notes

Shin splints are a nuisance for many endurance exercisers. This condition is often caused by overuse or tightness of the plantar flexor and invertor muscles. It is difficult to participate in sport activities when you're in constant pain from shin splints. This stretch particularly stretches the flexor digitorum longus and the medial aspect of the soleus muscles. People with this problem will definitely benefit from this stretch. Also evaluate shoes as well as the running and walking surfaces. In addition, a thorough stretching program should be included in any rehabilitation program.

It is more comfortable to do this stretch while wearing shoes. This is an excellent stretch for the flexor digitorum longus, medial soleus, and abductor hallucis muscles, which are located on the medial side of the lower leg and foot. Take extra care when placing the foot in an everted position, and be sure to progress slowly through the stretch. After the left heel reaches the floor or the lowest point, bending the left knee slightly can increase the stretch. This reduces the stretch on the hamstring muscles, but it increases the stretch on the flexor digitorum longus, medial soleus, and abductor hallucis muscles.

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